

SOHAM PARAB

sohamparab1003@gmail.com | +91-9307377878 | github.com/soham-03

Education

Fr. Conceicao Rodrigues College Of Engineering, Bandra

2022 - 2025

- Computer Engineering | CGPA: **9.29**

Vivekanand Education Society's Polytechnic, Chembur.

2019 - 2022

- Diploma in Computer Engineering, Aggregate: **93.14%**

Skills

C++ | C | Java | Python | Kotlin | Android Development | Jetpack Compose | K-Tor | React | Next | JavaScript | Firebase | Pocketbase | MongoDB | Unity

Work Experience

University of Mumbai | Software Development Intern

Jun'24 - Sep'24

- Developed a solution for the university using **Next.js** and **Pocketbase** to enhance data management for examination papers, student records, and other vital information
- User-based login for different types of users in the university

Shree Swami Samarth Pet Industries | App Development Intern

Sep'21 - Nov'21

- Made an Android App for the company using **Kotlin** and **XML** which is published on the Google Play Store
- Worked on the Backend API using the **K-tor** Framework
- Worked with **Firebase Firestore** for Chat Support and **Firebase Cloud Messaging** for notifications

Selected Projects

AI-Powered Virtual Reality Interview Simulator for DRDO (RAC)

Present

Developed an AI-powered Virtual Reality (VR) interview simulation platform tailored for DRDO's Recruitment and Assessment Centre (RAC)

- **3D AI-driven interviewer** with **emotional intelligence** and **real-time response analysis**
- **Actionable feedback** based on **role-specific questions**
- Integrated **emotion detection** and **attention span** measurement using **OpenCV**
- **Cheating detection** for unauthorized devices (e.g., **mobile phones**)
- Developed **"Talk to Resumes"** feature using **RAG architecture** for **candidate shortlisting**
- Added **role-based question prediction** and **YouTube video recommendations** for **candidate training**
- Designed **VR-based 360-degree Board Room tours** for a **real-life interview environment**

ChainEX: Blockchain-Based Foreign Money Transfer

Present

A blockchain-powered platform for secure, real-time foreign currency transfers with automatic tax compliance and fraud prevention

- **Real-time USD to INR** conversion using **Chainlink Price Feeds** and transparent blockchain-based transactions
- **Automatic tax calculation** and transfer to **government wallets** during currency exchange
- **Fraud prevention** with an ML model alerting users if **overcharging** occurs
- Personalized **itinerary suggestions** based on **location** and **customer organic rating**.

ArthVeda: Optimize Portfolio

June' 24

An Application and Website to optimize your investment portfolio. (made for TIAA retire-thon)

- An **Android Application** and a website made using **Django** which lets the user invest effectively and helps make an optimized portfolio
- A Portfolio Optimization Algorithm that used the **Monte Carlo Simulation** to get the Best Portfolio based on the user's preferred Rate of Risk, ROI, and Age
- **Analyze stocks with AI**, receive summarized stock news and data, and get assistance on portfolio decisions
- Leveraged **Yahoo Finance API** to get real-time stock prices
- **Portfolio Customization**: Pick your stocks, analyze, and visualize your portfolio, and customize it in the Portfolio Playground

Smart Farm: Increasing Precision Farming Using IOT and Machine Learning

March' 24

An Application that recommends crops based on various factors of soil and also helps to monitor the health of the field

- An Android Application with a clean and easy UI using **Jetpack Compose** and **MVVM** architecture
- Crop Recommendation ML model which uses **Naive Bayes** Classification and is Hosted using **FLASK**
- ML model which converts RGB Images to Hyperspectral Images using a **CNN** Model
- Hyperspectral Image Post-Processing using **MATLAB** to analyze the field

Ganesh Safety App for Fire and Security Association of India (FSAI)

Sep' 23

An Application that helped FSAI to register new mandals, conduct multiple audits, and directly get the results on the app

- An Android Application using **Kotlin** and **Jetpack Compose**, with **Firebase Firestore** as the Backend
- Stored the Latitude and Longitude of the registered mandals along with the image of the entrance, so that in case of a mishap the police can navigate easily

Academic and Extracurricular Achievements

- **Winner Technovate 2.0 2024** at **Sardar Patel Institute of Technology, Andheri**
- **Winner TIAA Retire-Thon 2024** at **TIAA Mumbai**
- **HackerRank Java 5 Star**
- **Runner-Up** in **Hack-Mania** Hackathon organized by IEEE and GDSC at **Atharva College of Engineering**
- **Third Position** in **Stratagem** Hackathon organized by GDA at **Fr. CRCE**
- **Ingenium Hackathon Runner-Up** organized by IEEE at **Fr. CRCE**

Publications

- **Winner Technovate 2.0 2024** at **Sardar Patel Institute of Technology, Andheri**
- **Winner TIAA Retire-Thon 2024** at **TIAA Mumbai**
- **HackerRank Java 5 Star**
- **Runner-Up** in **Hack-Mania** Hackathon organized by IEEE and GDSC at **Atharva College of Engineering**
- **Third Position** in **Stratagem** Hackathon organized by GDA at **Fr. CRCE**
- **Ingenium Hackathon Runner-Up** organized by IEEE at **Fr. CRCE**

Positions of Responsibility

Google Developer Student Clubs| Technical Advisor

Present

- Organized BitNBuild International hackathon collaborating with 14 National Universities and 14 Global Universities
- Managing the technical team and getting projects done within the deadlines

Google Developer Student Clubs| Mobile Application Development Lead

July' 23 - July' 24

- Working on various projects for the College
- Helped students in the college with mobile application development by conducting workshops